Indian Institute of Technology Bhubaneswar

School of Infrastructure

Session: Spring 2026

Finite Element Method (CE6L314) — Class Test 1

Date: February 21, 2026 — Total Marks: 50

Instructions:
(1) All questions are mandatory.
(2) Assume a reasonable value of any missing data.
(3) Provide neatly drawn figures whenever needed.

1. Consider the boundary value problem

$$-u''(x) = f(x), \quad x \in (0,1)$$

$$u(0) = 0, \quad u(1) = 0$$

(a) Show that the problem is equivalent to minimizing a functional.
(b) Derive the energy functional $J(u)$.
(c) Prove that the Euler–Lagrange equation of $J(u)$ gives the original differential equation.
(d) Specify the admissible function space. [10]

2. Consider

$$-u'' = 1, \quad x \in (0,1)$$

$$u(0) = u(1) = 0$$

Assume the trial function

$$u_N(x) = a_1 x(1-x)$$

(a) Write the energy functional.
(b) Substitute the trial function into the functional.
(c) Minimize with respect to a_1.
(d) Compare with the exact solution. [5]

3. Consider

$$-u''(x) = f(x), \quad x \in (0,1)$$

$$u(0) = 0, \quad u'(1) = g$$

   (a) Derive the weak form.
   (b) Show how the Neumann boundary condition appears naturally.
   (c) Explain why the boundary condition $\boxed{u'(1) = g}$ is called a natural boundary condition. [5]

4. For an element, $\Omega^{(e)} = [x_i, x_{i+1}]$:
   (a) Derive quadratic shape functions $N_1^{(e)}$, $N_2^{(e)}$ and $N_3^{(e)}$.
   (b) Show the partition of unity property.
   (c) Why do the shape functions satisfy the partition of unity property? [10]

5. Consider

$$-u'' = 1, \quad x \in (0,1)$$

$$u(0) = 0, \quad u(1) = 0$$

   Using a uniform mesh with two and three equal elements:
   (a) Construct the global hat basis functions.
   (b) Write the finite-dimensional approximation space.
   (c) Derive the stiffness matrix.
   (d) Derive the load vector.
   (e) Solve the resulting algebraic system.
   (f) Compare with the exact solution. [10]

6. Consider

$$-u'' = 1, \quad x \in (0,1)$$

$$u(0) = 0, \quad u(1) = 0$$

   Using non-uniform mesh:

$$x_0 = 0, \quad x_1 = 0.3, \quad x_2 = 1$$

   (a) Construct the global hat functions.
   (b) Starting from the weak form, derive the algebraic system of equations in matrix-vector form.
   (c) Derive stiffness matrix entries explicitly. [10]